Institute of Information Management

National Yang Ming Chiao Tung University

Master Thesis

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

Student: Hsu, Pei-Chih

Advisor: Liu, Duen-Ren

June 2026

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

Student: Pei-Chih Hsu
Advisor: Dr. Duen-Ren Liu

A Thesis
Submitted to Institute of Information Management
College of Management
National Yang Ming Chiao Tung University
in partial Fulfillment of the Requirements
for the Degree of
Master of Science
in
Information Management

June 2026
Taiwan

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

Student : Pei-Chih Hsu   Advisor: Dr. Duen-Ren Liu

Institute of Information Management
National Yang Ming Chiao Tung University

Abstract

Knowledge-graph-aware recommenders often improve accuracy when the KG is dense and reliable. However, in review-derived, cold-start, or privacy-constrained settings, KGs are often sparse or noisy, and this regime remains underexplored. Using a sparse review-derived product KG with only 2.4 aspect edges per item after filtering, we observe that representative KG-aware methods are all outperformed by a plain collaborative-filtering model that ignores the KG entirely. The main reason is that these methods hardwire the KG signal into the scoring pipeline, with no mechanism to reduce its influence when the KG is unreliable, allowing KG noise to interfere with collaborative-signal learning.

We propose RA-GARK, which treats the KG as a controllable side channel rather than a mandatory component of the scoring process, realized in three steps. First, each item's graph content is summarized into a few representative aspects and initialized from semantic vectors instead of purely random values. Second, the model automatically selects the most relevant aspect for each user and item. Third, a fusion gate decides whether to use this side channel: it starts almost closed, so training begins from pure collaborative filtering, and opens only when the side channel is useful, allowing the model to fall back gracefully when it is not. Without altering the KG itself, RA-GARK turns the KG's contribution from negative to positive.

This work shows that the poor performance of KG-aware recommenders on sparse KGs is not because the KG contains no useful information, but because existing architectures lack the ability to use the KG selectively. The same KG can be a liability under mandatory fusion, yet become an asset once the model can decide whether to use it. This kind of graceful degradation is not specific to knowledge graphs; it is also valuable in any setting that combines a reliable primary signal with an auxiliary signal of uncertain quality, such as cold-start, multimodal, or cross-domain recommendation.

Keywords: knowledge-graph-aware recommendation; graceful degradation; fusion gate; sparse knowledge graph; collaborative filtering

Contents

List of Figures

List of Tables

1 Introduction

1.1 Background

Graph neural networks (GNNs) are now a standard backbone for collaborative filtering. LightGCN [1] showed that the feature transformations and nonlinearities used in earlier graph recommenders such as NGCF [2] are not necessary for strong ranking performance: linear neighborhood aggregation over the user–item graph, followed by layer-wise averaging, is often enough. Its simplicity makes it a strong default, and later self-supervised methods such as SGL [3] build on the same backbone.

Knowledge-graph-aware recommendation augments collaborative signals with item-side semantics. KGAT [4] propagates over a unified user–item–entity graph; KGCL [5] and MCCLK [6] align KG and collaborative views through contrastive learning; KGRec [7] further introduces edge-level rationales. These methods are effective when the KG is dense and reliable. This thesis studies the less favorable, but common, setting in which the KG is sparse and noisy.

1.2 An Empirical Anomaly under Sparse KG

Our starting point is an empirical anomaly. On the review-derived Amazon Books KG used in this thesis, with only 2.4 aspect edges per item after filtering, all four representative KG-aware methods fall below pure LightGCN. Table 1 reports the NDCG@20 comparison.

Table 1: Comparison of KG-aware methods and pure LightGCN on NDCG@20.

Method	NDCG@20
MCCLK [6]	0.1067
KGCL [5]	0.1073
KGAT [4]	0.1079
KGRec [7]	0.1095
Pure LightGCN [1]	**0.1179**

The result does not imply that these baselines are generally weak; they perform well on the dense KG benchmarks used in their original papers. Rather, it exposes a shared failure mode under sparse KG: when KG signal is directly fused into the scoring path, KG noise can enter the same gradient flow that produces the collaborative representation.

1.3 Sparse KG as a Practical Setting

Sparse KG is not merely a dataset accident. Review-derived KGs inherit their coverage from what users mention, so many items receive few or uneven aspect tags. Cold-start domains, low-resource languages, and niche product verticals often lack curated sources comparable to Freebase or Wikidata. Privacy-constrained domains may also restrict which relational signals can be exposed. In these settings, KG completion is not a guaranteed remedy because it needs seed signal and can introduce additional noise.

For this reason, robustness when side information is unreliable is as important as peak performance when side information is clean. This thesis takes sparse-KG robustness as the main design objective.

1.4 Research Questions

The empirical anomaly motivates two questions:

- **RQ1**: Why do existing KG-aware methods underperform pure LightGCN under sparse KG?

- **RQ2**: What architecture can exploit KG signal when useful while preventing KG noise from contaminating collaborative filtering?

We answer RQ1 by identifying mandatory KG fusion as the shared structural weakness. We answer RQ2 with RA-GARK, which treats the KG as a gateable side channel rather than as a required part of message passing. The model starts close to pure LightGCN and learns to open the KG channel only when doing so reduces ranking loss.

1.5 Thesis Organization

Chapter 2 reviews graph collaborative filtering, KG-aware recommendation, gating mechanisms, and attention normalization. Chapter 3 presents RA-GARK: the LightGCN local view, KG-SVD aspect-slot initialization, softmax rationale masking, local-biased fusion gate, contrastive regularization, and training procedure. Chapter 4 reports results, ablations, sensitivity analyses, stability checks, a case study, and cost. Chapter 5 summarizes the findings, limitations, and future work.

1.6 Contributions

This thesis makes four contributions:

1. It documents a sparse-KG failure case in which KGAT, KGCL, MCCLK, and KGRec all underperform pure LightGCN.
2. It proposes RA-GARK, a dual-view recommender that integrates KG information through a gateable side channel instead of mandatory fusion.
3. It shows that RA-GARK outperforms both the strongest KG-aware baseline and pure LightGCN on the same sparse KG.
4. It reports a methodological observation on attention normalization: in this setting, softmax rationale masking is much more stable than sigmoid masking, suggesting that normalization should match the semantics and scale constraints of the auxiliary channel.

2 Related Work

This chapter reviews the work most relevant to RA-GARK: graph collaborative filtering, KG-aware recommendation, rationale-aware KGRec, gating mechanisms, and attention normalization.

2.1 Foundations and KG-Aware Baselines

LightGCN [1] is the immediate predecessor of RA-GARK's local view. It removes the feature transformation and nonlinear activation used by NGCF [2], retaining only normalized propagation over the user–item bipartite graph and layer-wise embedding averaging. The result is simple, stable, and strong; later models such as SGL [3] and DCCF [8] add self-supervised objectives on top of a LightGCN-style backbone.

RA-GARK adopts LightGCN unchanged for the local view. This choice is deliberate: on our sparse review-derived KG, pure LightGCN already outperforms all evaluated KG-aware baselines, so any KG module must improve over this default without degrading the collaborative signal.

KGAT [4] is the canonical deep-fusion KG recommender. It merges user–item interactions and KG triples into a collaborative knowledge graph, then propagates messages over the unified structure. This allows KG entities to directly shape user and item representations. The same idea appears, with different regularization, in later KG-aware methods. The weakness under sparse KG is that the KG cannot be removed from the propagation path when it is noisy.

2.1.1 Contrastive Knowledge-Graph Methods

KGCL [5] perturbs the KG and aligns original and perturbed views with an InfoNCE [9] objective. **MCCLK** [6] extends this idea to collaborative, semantic, and structural views. Both methods assume that KG-derived views contain enough reliable structure for contrastive learning to regularize the recommender. In our sparse setting, the semantic and structural views are too weak, so contrastive alignment can amplify noise rather than useful invariance.

RA-GARK also uses contrastive learning, but only as a small auxiliary regularizer. The losses align local and global geometry with weight $\lambda_{\mathrm{CL}} = 0.005$ and stop-gradient on the KG side. Fusion itself is handled by a learned gate, not by contrastive agreement.

2.1.2 Rationale-Aware Recommendation: KGRec

KGRec [7] is closest to RA-GARK in motivation because it explicitly models rationales. It scores KG edges, applies Bernoulli dropout to selected edges, and uses contrastive learning to make recommendations robust to rationale removal. However, KGRec's rationale module operates inside a KGAT-style message-passing path. Thus, even though individual edges are weighted, the KG as a whole remains mandatory.

RA-GARK adopts the rationale idea but changes its site of action. It selects among latent item-aspect slots in a side channel, then lets a local-biased fusion gate decide how much of that channel enters the final score.

2.2 Rationale Paradigm: Shared Premise vs. Diverged Trust Assumption

KGRec and RA-GARK share the premise that not all KG evidence is equally useful. They differ in their trust assumption. KGRec assumes that useful edges exist and should be identified within the KG propagation path. RA-GARK makes the weaker assumption that the entire KG may be unreliable; therefore, the model must be able to disengage the KG channel.

- **Rationale granularity**: KGRec selects KG triples; RA-GARK selects among $A = 4$ latent aspect slots initialized from an IDF-weighted item–aspect matrix.
- **Selection mechanism**: KGRec uses Bernoulli dropout; RA-GARK uses differentiable softmax attention.
- **Site of action**: KGRec applies rationalization inside message passing; RA-GARK applies it in a separate global view.
- **KG trust**: KGRec has no mechanism to shut off the KG; RA-GARK starts near pure LightGCN with $\alpha^{(0)} \approx 0.993$.

Table 2: Comparison of rationale paradigms between KGRec and RA-GARK.

Axis	KGRec	RA-GARK
Rationale granularity	KG triple (edge)	A latent aspect slots
Selection mechanism	Bernoulli dropout + CL	Softmax attention
Site of action	Inside KGAT propagation	Side channel, late fusion
Treatment of KG trust	No disengage mechanism	Bias-initialized gate

2.3 Gating Mechanisms and the Fusion-Gate Gap

RA-GARK's fusion gate follows a broad design pattern from deep learning: bias the model toward a reliable default, then let training open an additional path when useful. Highway Networks [10] use a transform gate initialized toward the identity path, allowing deep networks to train without forcing every layer to transform the signal. RA-GARK applies the same idea to recommendation: LightGCN is the reliable default, and the KG channel is opened only if it reduces loss.

Mixture-of-Experts methods such as MMoE [11] and PLE [12] also use gates, but their experts are typically symmetric candidates. RA-GARK's setting is asymmetric: the collaborative branch is empirically reliable, while the KG branch may be harmful. This asymmetry motivates a biased gate rather than a neutral mixture.

2.3.1 No Fusion Gate in KG-Aware Recommendation

The KG-aware recommenders we survey integrate KG signal through propagation, contrastive alignment, attention, cross-feature units, or intent disentanglement. None provides a bias-initialized gate that can architecturally disengage the KG channel under sparse or unreliable KG.

Table 3: KG integration mechanisms in representative recommenders.

Method	KG integration mechanism	Disengageable?
KGAT [4]	Direct entity-message propagation	No
KGCL [5]	Perturbed-KG contrastive learning	No
MCCLK [6]	Multi-view contrastive alignment	No
KGRec [7]	Edge-level dropout + KGAT aggregator	No
CKAN [13]	Collaborative-knowledge attention	No
MKR [14]	Cross-feature transfer unit	No
KGIN [15]	Intent disentanglement over KG	No
RA-GARK (this work)	**Bias-initialized fusion gate**	**Yes**

Table 4: Design comparison of KG-aware recommendation methods.

Method	Local aggregator	KG integration	Rationale
KGAT	Bi-interaction	Direct propagation	None
KGCL	KGAT-style	Propagation + CL	None
MCCLK	Multi-view	Multi-view CL	None
KGRec	KGAT-style	Propagation + edge dropout	Edge Bernoulli
RA-GARK	**Pure LightGCN**	**Gated late fusion**	**Aspect softmax**

The claim is intentionally narrow: among the methods examined here, none uses a bias-initialized fusion gate to provide graceful degradation when KG quality is low.

2.4 Attention Normalization

Rationale selection also depends on the normalization used by the attention head. DIN [16] uses sigmoid attention over historical behaviors, treating each behavior as independently relevant. NAIS [17] and AFM [18] use softmax over competing historical items or feature interactions. The correct choice depends on the semantic assumption.

RA-GARK uses softmax because its latent slots are competing explanations of the same item's KG semantics. In Chapter 4, replacing softmax with sigmoid produces the largest ablation loss. We interpret this not as a universal rule, but as evidence that normalization must match both the rationale semantics and the scale constraints imposed by the downstream fusion gate.

2.5 Summary and Positioning of RA-GARK

RA-GARK is not positioned as a strict replacement for all KG-aware recommenders. It occupies a sparse-KG design point: the KG is useful enough to try, but unreliable enough that the model needs an explicit way to ignore it.

3 Proposed Method

This chapter presents RA-GARK. Section 3.1 gives the design principle and problem setup; Sections 3.2–3.5.1 define the local view, KG-derived global view, rationale masking, fusion gate, and contrastive regularization; Section 3.6 summarizes training and cost.

3.1 Design Principle and Problem Setup

The failure mode identified in Chapter 1 is mandatory KG fusion. Existing KG-aware baselines inject KG embeddings into the scoring path and provide no architectural switch for ignoring KG signal when it is unreliable. RA-GARK instead follows one principle:

> *The KG should be a side channel whose contribution is learned by a gate, initialized toward pure collaborative filtering and opened only when the KG reduces loss.*

This principle yields three design choices. First, RA-GARK has two disjoint views: a LightGCN local view $(u_{\mathrm{loc}}, i_{\mathrm{loc}})$ over the user–item graph and a KG-derived global view $(u_{\mathrm{glo}}, i_{\mathrm{glo}})$. Second, the views are combined only at the final scoring stage by user- and item-side gates. The gate bias is initialized so that $\alpha^{(0)} \approx 0.993$, making the initial model almost pure LightGCN. Third, the KG item representation is stored in $A = 4$ latent aspect slots initialized by truncated SVD of an IDF-weighted item–aspect matrix and selected by softmax rationale masking.

3.1.1 Architectural Overview

Figure 1 shows the inference flow. The local lane applies LightGCN to obtain collaborative embeddings. The global lane looks up u_{glo} and forms i_{glo} by applying softmax weights over the item's SVD-initialized aspect slots. Fusion gates mix the two views, and the final score is an inner product. The KG-SVD step is an initialization procedure, not a forward-pass operation.

3.1.2 Notation

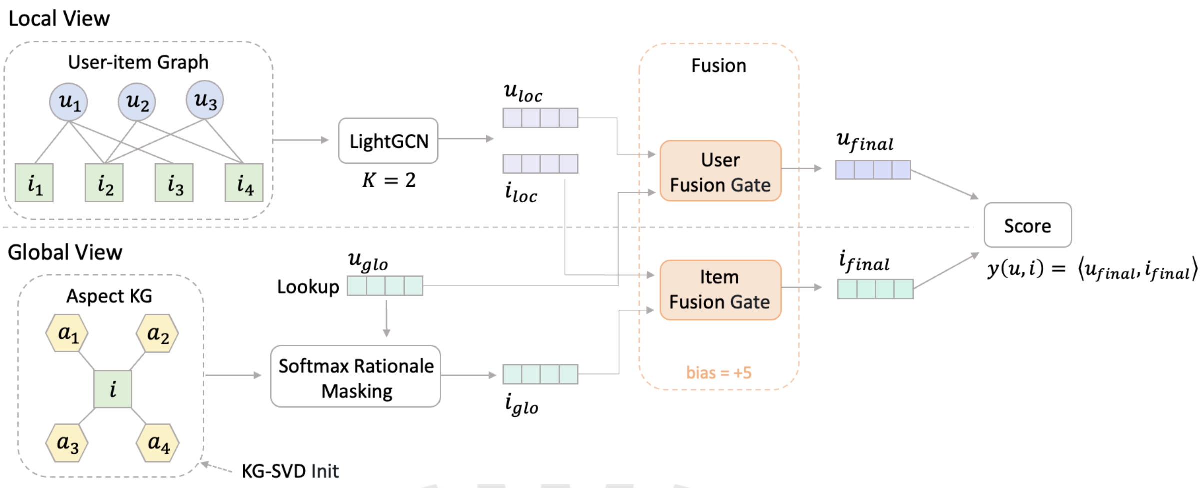

Figure 1: RA-GARK architecture.

Table 5: Notation used in the RA-GARK formulation.

Symbol	Meaning
$\mathcal{U}, \mathcal{I}$	sets of users and items
$\mathcal{R} \subseteq \mathcal{U} \times \mathcal{I}$	observed user–item interactions
$\mathcal{G}_{\text{KG}}$	item–aspect knowledge graph
$\mathcal{A}$	set of aspect tags
A	number of latent aspect slots per item ($A = 4$)
d	embedding dimension ($d = 128$ unless stated otherwise)
K	number of LightGCN layers ($K = 2$)
$u_{\text{loc}}, i_{\text{loc}}$	local-view embeddings
u_{glo}	global-view user embedding
$\mathbf{a}_i \in \mathbb{R}^{A \times d}$	latent aspect slots of item i
i_{glo}	global-view item embedding
α_u, α_i	user- and item-side fusion gate outputs
$\hat{y}(u, i)$	predicted preference score

3.1.3 Problem Formulation

We study implicit-feedback top-K recommendation. Given users, items, observed interactions, and an item–aspect KG, the model ranks unseen items for each user and is evaluated by NDCG, Recall, Hit Rate, and MAP.

RA-GARK scores a pair (u, i) by

$$\hat{y}(u, i) = \langle u_{\text{final}},\ i_{\text{final}} \rangle, \tag{1}$$

where

$$u_{\text{final}} = \alpha_u\, u_{\text{loc}} + (1 - \alpha_u)\, u_{\text{glo}}, \qquad i_{\text{final}} = \alpha_i\, i_{\text{loc}} + (1 - \alpha_i)\, i_{\text{glo}}. \tag{2}$$

Training minimizes BPR plus two auxiliary contrastive losses:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \lambda_{\mathrm{CL}}\big(\mathcal{L}_{\mathrm{aCL}} + \mathcal{L}_{\mathrm{uCL}}\big), \tag{3}$$

$$\mathcal{L}_{\mathrm{BPR}} = -\log\sigma\big(\hat{y}(u,i^{+}) - \hat{y}(u,i^{-})\big). \tag{4}$$

The contrastive terms align local and global geometry with stop-gradient on the global side, and λ_{CL} is fixed at 0.005.

3.2 Local View: LightGCN

The local view is exactly LightGCN [1]. Let $\mathbf{R}$ be the binary interaction matrix. The bipartite adjacency and normalized adjacency are

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^{\top} & \mathbf{0} \end{bmatrix}, \tag{5}$$

$$\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\,\mathbf{A}\,\mathbf{D}^{-1/2}, \qquad \mathbf{D} = \mathrm{diag}(\mathbf{A}\,\mathbf{1})\,. \tag{6}$$

Starting from trainable user and item embeddings $\mathbf{E}^{(0)}$, propagation is

$$\mathbf{E}^{(\ell+1)} = \tilde{\mathbf{A}}\,\mathbf{E}^{(\ell)}, \qquad \ell = 0, 1, \ldots, K-1. \tag{7}$$

The final local table is the layer-wise average

$$\bar{\mathbf{E}} = \frac{1}{K+1}\sum_{\ell=0}^{K}\mathbf{E}^{(\ell)}. \tag{8}$$

We use $K = 2$. No KG operation enters this branch, so u_{loc} and i_{loc} remain purely collaborative.

3.3 Global View: KG-SVD Initialization

The global view stores item KG semantics in a learnable tensor

$$\mathbf{a}_i \in \mathbb{R}^{A\times d}, \qquad A = 4, \tag{9}$$

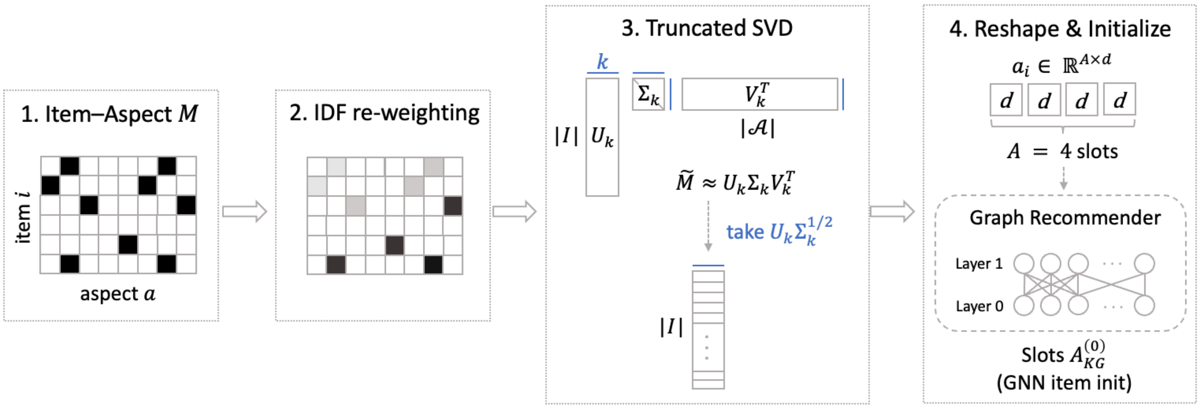

Figure 2: KG-SVD initialization of item aspect slots.

collected over all items as $\mathbf{A}_{\text{KG}} \in \mathbb{R}^{|\mathcal{I}| \times A \times d}$. The slots are latent directions, not raw aspect tags. They are initialized once by KG-SVD and then updated by gradient descent.

We first build the binary item–aspect matrix

$$\mathbf{M}_{i,a} = \begin{cases} 1, & (i,a) \in \mathcal{G}_{\text{KG}}, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Frequent generic aspects are downweighted by IDF:

$$\text{idf}(a) = \log\left(\frac{|\mathcal{I}|}{|\{i : \mathbf{M}_{i,a} = 1\}| + 1}\right) + 1, \tag{11}$$

$$\tilde{\mathbf{M}}_{i,a} = \text{idf}(a) \cdot \mathbf{M}_{i,a}. \tag{12}$$

Then we compute the rank-k SVD with $k = Ad$:

$$\tilde{\mathbf{M}} \;\approx\; \mathbf{U}_k \, \mathbf{\Sigma}_k \, \mathbf{V}_k^{\top}, \qquad \mathbf{U}_k \in \mathbb{R}^{|\mathcal{I}| \times k}, \tag{13}$$

and form

$$\mathbf{E}_{\text{KG}} = \mathbf{U}_k \, \mathbf{\Sigma}_k^{1/2}. \tag{14}$$

Finally, the flat item embedding is reshaped into slots:

$$\mathbf{A}_{\text{KG}}^{(0)}[i,k,:] = \mathbf{E}_{\text{KG}}[i, kd : (k+1)d], \qquad k = 0, \ldots, A-1. \tag{15}$$

The tensor is rescaled to the Xavier-normal target standard deviation $\sqrt{2/(A+d)}$ so that KG slots start on a magnitude compatible with the other model parameters. If the aspect vocabulary were smaller than Ad, missing dimensions would be filled with small Gaussian noise, but this fallback is not used in our experiments.

The ablation in Section 4.4 shows that replacing KG-SVD with random initialization costs 6.0% NDCG@20, indicating that the sparse KG does contain useful geometry but is hard to recover from random parameters.

3.4 Global View: Softmax Rationale Masking

For each pair (u, i), the rationale module chooses how to combine the A slots of item i. The global user embedding is a separate lookup:

$$u_{\text{glo}} = \mathbf{E}^{\text{glo}}_u, \qquad \mathbf{E}^{\text{glo}} \in \mathbb{R}^{|\mathcal{U}|\times d}. \tag{16}$$

For each slot k, the attention logit is

$$\ell_{u,i,k} = \text{MLP}([u_{\text{glo}} \,\|\, \mathbf{a}_{i,k}])\,, \tag{17}$$

with

$$\text{MLP}(\mathbf{z}) = \mathbf{w}^\top \cdot \text{LeakyReLU}(\mathbf{W}\,\mathbf{z})\,. \tag{18}$$

The same MLP is shared across slots.

The logits are normalized by a temperature-scaled softmax:

$$w_{u,i,k} = \frac{\exp(\ell_{u,i,k}/\tau)}{\sum_{k'=1}^{A}\exp(\ell_{u,i,k'}/\tau)}. \tag{19}$$

We use $\tau = 0.5$. The global item embedding is

$$i_{\text{glo}} = \sum_{k=1}^{A} w_{u,i,k}\cdot \mathbf{a}_{i,k}. \tag{20}$$

Softmax is important because it keeps $\sum_k w_{u,i,k} = 1$, bounding the scale of i_{glo}. A sigmoid head treats slots independently and lets the embedding magnitude drift with the logits. In Section 4.4, the sigmoid variant causes the largest ablation loss. The case study later shows that the learned softmax weights are softly peaked rather than one-hot; the useful property is magnitude control for a throttled KG channel.

3.5 Local-Biased Fusion and Contrastive Regularization

The fusion gate implements the side-channel design. RA-GARK uses separate scalar gates for users and items:

$$\alpha_u = \mathrm{Gate}_u([u_{\mathrm{loc}} \,\|\, u_{\mathrm{glo}}])\,, \tag{21}$$

$$\alpha_i = \mathrm{Gate}_i([i_{\mathrm{loc}} \,\|\, i_{\mathrm{glo}}])\,. \tag{22}$$

The gates have independent parameters because the informativeness of KG signal differs by side.

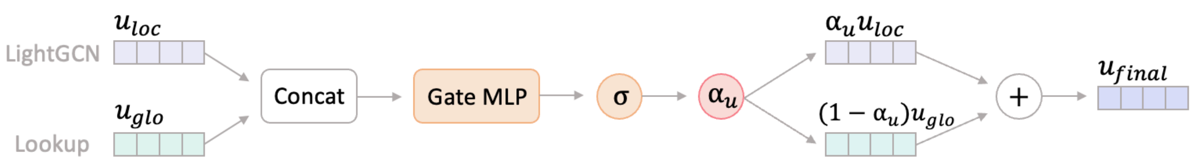

Figure 3: Local-biased fusion gate for combining local and global views.

Each gate is a small MLP with a sigmoid head:

$$\mathrm{Gate}(\mathbf{z}) = \sigma\big(\mathbf{w}^\top \tanh(\mathbf{W}\mathbf{z}) + b\big)\,. \tag{23}$$

The key initialization is $b = +5$, giving

$$\alpha^{(0)} \approx \sigma(b) = \sigma(5) \approx 0.993. \tag{24}$$

Thus the initial mixture is

$$u^{(0)}_{\mathrm{final}} \approx 0.993\, u^{(0)}_{\mathrm{loc}} + 0.007\, u^{(0)}_{\mathrm{glo}}, \tag{25}$$

and similarly on the item side. The model starts as almost pure LightGCN; gradients must push the gate open before the KG can materially affect the score. A smaller bias exposes training to KG noise, while a much larger bias makes the gate difficult to move. The sweep in Section 4.5.3 supports $b = +5$.

3.5.1 Cross-View Contrastive Regularization

Because the two views live in different signal spaces, RA-GARK adds small contrastive losses to align their angular geometry. These losses are auxiliary; the fusion gate remains the integration mechanism.

For item slots,

$$\mathcal{L}_{\mathrm{aCL}} = \frac{1}{A}\sum_{k=1}^{A} \mathrm{InfoNCE}(g(i_{\mathrm{loc}}),\ \mathrm{sg}[\mathbf{a}_{i,k}])\,, \tag{26}$$

where g is a projection head and sg is stop-gradient. InfoNCE is

$$\mathrm{InfoNCE}(\mathbf{x}, \mathbf{y}) = -\log \frac{\exp(\langle \mathbf{x}, \mathbf{y}\rangle/\tau_{\mathrm{CL}})}{\sum_{j=1}^{B} \exp(\langle \mathbf{x}, \mathbf{y}_j\rangle/\tau_{\mathrm{CL}})}, \tag{27}$$

with $\tau_{\mathrm{CL}} = 0.2$. The user loss is

$$\mathcal{L}_{\mathrm{uCL}} = \mathrm{InfoNCE}(g(u_{\mathrm{loc}}),\ \mathrm{sg}[u_{\mathrm{glo}}])\,. \tag{28}$$

Stop-gradient prevents the contrastive objective from overwriting the SVD-initialized KG geometry; the local view absorbs the alignment cost.

3.6 Training and Computational Complexity

3.6.1 Total Objective

The full objective is Eq. (3): BPR provides the ranking signal, and the two contrastive terms provide weak cross-view alignment.

3.6.2 Optimization

We train with Adam, learning rate 10^{-3}, batch size 128, for up to 80 epochs. Early stopping uses validation NDCG@20 with patience 10. Each batch contains triplets (u, i^+, i^-), with i^- sampled uniformly from unobserved items for user u.

3.6.3 Inference

At inference time, LightGCN propagation is computed once and cached. For each user, item scores are computed in a vectorized pass over all candidate items, including rationale masking, item gating, and the final inner product.

3.6.4 Computational Complexity

Table 6: Per-batch training cost of RA-GARK modules.

Module	Cost
LightGCN propagation	$O(\lvert\mathcal{E}_{\mathcal{UI}}\rvert\, d\, K)$
u_{glo} lookup	$O(B\, d)$
Aspect-slot read	$O(B\, A\, d)$
Rationale masking	$O(B\, A\, d)$
User and item fusion gates	$O(B\, d^2)$
BPR + contrastive losses	$O(B\, d)$

The dominant term is LightGCN propagation, as in the baseline. The KG additions are linear in A and d except for the small gate MLPs, and remain a small constant-factor overhead at $A = 4$.

4 Experiments and Results

This chapter evaluates RA-GARK on the sparse review-derived KG introduced earlier. It reports the dataset, protocol, baselines, main ranking results, ablations, sensitivity checks, a case study, and computational cost.

4.1 Dataset and KG Construction

4.1.1 Dataset

We use a filtered Amazon Books subset represented by user–item interactions and an item–aspect KG. After preprocessing, the dataset contains 905 users, $1{,}399$ items, $22{,}265$ interactions, and $3{,}370$ KG edges over $2{,}098$ aspect tags. Each item has only 2.4 aspect edges on average, which places the benchmark in the sparse-KG regime.

Table 7: Dataset statistics after preprocessing.

Statistic	Value
#Users	905
#Items	1,399
#Interactions	22,265
#KG edges (after filtering)	3,370
#Unique aspects	2,098
Avg. edges per item	2.4
Interaction density	1.76%

Interactions are split user-stratified $70/15/15$ for train, validation, and test. The split seed is fixed at 42 and reused across all baselines.

4.1.2 Knowledge Graph Construction

The item–aspect KG follows the pipeline of [19]. Product reviews and cover images are summarized with visual and textual encoders, and aspect mentions are extracted as $(\mathit{item}, \texttt{has\_aspect}, \mathit{asp}$ triples. From $13{,}905$ raw candidate edges, we remove the top 2% most frequent generic aspects and a small stopword list of domain-generic tags. The resulting $3{,}370$ edges are used unchanged by all KG-aware methods.

4.2 Experimental Setup

4.2.1 Baselines

We compare against pure LightGCN [1] and four KG-aware baselines: KGAT [4], KGCL [5], MCCLK [6], and KGRec [7]. These cover deep KG propagation, contrastive KG alignment, multi-view contrastive learning, and edge-level rationale learning. Baselines are reproduced from released code on the same processed KG and split. We keep published defaults where possible, align the embedding dimension to $d = 128$, and use the same epoch budget and early-stopping patience.

4.2.2 Evaluation Protocol

We use full ranking: for each test user, every item except the user's training items is scored and ranked. Sampled negative evaluation is not used. NDCG@20 is the primary metric; Recall, Hit Rate, and MAP are secondary. Early stopping uses validation NDCG@20, and reported test metrics come from the best validation epoch.

4.2.3 Hyperparameter Settings

Table 8: RA-GARK hyperparameter settings used in experiments.

Hyperparameter	Value
Embedding dimension d	128
LightGCN layers K	2
Aspect slots A	4
Rationale temperature τ	0.5
Fusion-gate bias b (init)	$+5$
Contrastive weight λ_{CL}	0.005
InfoNCE temperature τ_{CL}	0.2
Optimizer	Adam (β_1=0.9, β_2=0.999)
Learning rate	10^{-3}
Batch size	128
Max epochs	80
Early-stopping patience	10
Random seed	42

4.2.4 Hardware and Software

Experiments run on a single NVIDIA GeForce RTX 3090 GPU with 24 GB memory. The implementation uses PyTorch 2.6.0 with CUDA 12.6 and FP32 training.

4.3 Main Results

Tables 9 and 10 show that RA-GARK is strongest at both cutoffs. At Top-20 it reaches NDCG@20 $= 0.1243$, improving over KGRec by $+13.5\%$ and over pure LightGCN by $+5.4\%$. This is the central result: the same sparse KG that hurts all mandatory-fusion baselines becomes beneficial when routed through a gateable side channel.

Table 9: Comparison of recommendation performance under Top-20 evaluation.

Model	NDCG@20	HR@20	Recall@20	MAP@20
MCCLK [6]	0.1067	0.4530	0.1720	0.0497
KGCL [5]	0.1073	0.4696	0.1827	0.0479
KGAT [4]	0.1079	0.4773	0.1807	0.0491
KGRec [7]	0.1095	0.4729	0.1834	0.0500
LightGCN [1]	0.1179	0.4917	0.1937	0.0555
RA-GARK	**0.1243**	**0.4972**	**0.2020**	**0.0594**
Δ vs KGRec	+13.5%	+5.1%	+10.1%	+18.8%
Δ vs LightGCN	+5.4%	+1.1%	+4.3%	+7.0%

Table 10: Comparison of recommendation performance under Top-10 evaluation.

Model	NDCG@10	HR@10	Recall@10	MAP@10
MCCLK [6]	0.0804	0.3182	0.1047	0.0416
KGCL [5]	0.0809	0.3260	0.1096	0.0410
KGAT [4]	0.0786	0.3215	0.1102	0.0388
KGRec [7]	0.0874	0.3249	0.1155	0.0465
LightGCN [1]	0.0908	0.3436	0.1201	0.0483
RA-GARK	**0.0966**	**0.3558**	**0.1265**	**0.0520**
Δ vs KGRec	+10.5%	+9.5%	+9.5%	+11.8%
Δ vs LightGCN	+6.4%	+3.6%	+5.4%	+7.7%

The Top-10 results preserve the same ordering, showing that the gain is not an artifact of deeper recommendation lists.

4.4 Ablation Study

We remove one component at a time and retrain with all other settings fixed. The ablations test core components, auxiliary contrastive losses, and diagnostic variants.

Table 11: Ablation results for RA-GARK design components under Top-20 evaluation.

Model	NDCG@20	MAP@20
RA-GARK (full)	$\mathbf{0.1243}$	$\mathbf{0.0594}$
w/o softmax head	0.1005	0.0451
w/o KG-SVD init	0.1171	0.0545
w/o fusion-gate bias	0.1194	0.0555
w/o MLP gate	0.1180	0.0552
w/o user CL ($\mathcal{L}_{\mathrm{uCL}}$)	0.1192	0.0563
w/o aspect CL ($\mathcal{L}_{\mathrm{aCL}}$)	0.1200	0.0570
w/o rationale-enabled selection	0.1213	0.0568
w/o global view (CL-only dual view)	0.1219	0.0575

Table 12: Ablation results for RA-GARK design components under Top-10 evaluation.

Model	NDCG@10	MAP@10
RA-GARK (full)	$\mathbf{0.0960}$	$\mathbf{0.0519}$
w/o softmax head	0.0785	0.0397
w/o KG-SVD init	0.0922	0.0479
w/o fusion-gate bias	0.0923	0.0482
w/o MLP gate	0.0926	0.0484
w/o user CL ($\mathcal{L}_{\mathrm{uCL}}$)	0.0924	0.0492
w/o aspect CL ($\mathcal{L}_{\mathrm{aCL}}$)	0.0940	0.0502
w/o rationale-enabled selection	0.0943	0.0495
w/o global view (CL-only dual view)	0.0949	0.0504

The largest loss comes from replacing the softmax head, confirming that attention normalization is central in this architecture. KG-SVD initialization and local-biased gating also produce clear gains. Contrastive losses help, but their smaller effect matches their intended auxiliary role.

4.5 Sensitivity Analysis

We sweep four hyperparameters: rationale temperature τ, slot count A, fusion bias b, and contrastive weight λ_{CL}. Figure 4 summarizes the trends.

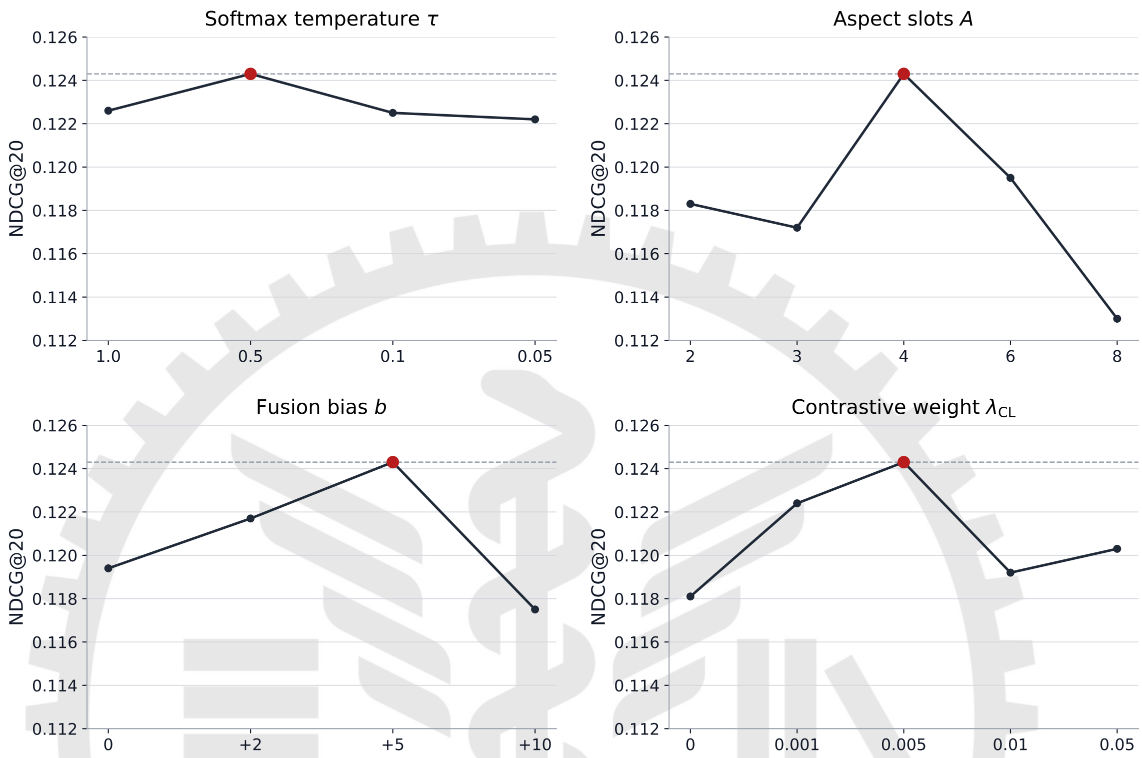

Figure 4: Sensitivity of RA-GARK to the four main hyperparameters under NDCG@20.

4.5.1 Softmax Temperature τ

The default $\tau = 0.5$ is best among $\{0.05, 0.1, 0.5, 1.0\}$. Lower temperatures over-sharpen the slot distribution, while $\tau = 1.0$ flattens it toward a near-uniform slot average.

4.5.2 Number of Aspect Slots A

The default $A = 4$ performs best among $\{2, 3, 4, 6, 8\}$. Smaller values under-allocate the latent KG space; larger values dilute already sparse evidence across too many slots.

4.5.3 Fusion-Gate Bias b

The bias sweep over $\{0, +2, +5, +10\}$ validates the local-biased design. Without bias, NDCG@20 drops to 0.1196; with $b = +10$, the gate is too saturated to adapt. The default

$b = +5$ is the best tradeoff between a near-LightGCN start and non-vanishing gradients.

4.5.4 Contrastive Weight λ_{CL}

The default $\lambda_{\mathrm{CL}} = 0.005$ is best among $\{0, 0.001, 0.005, 0.01, 0.05\}$. Removing contrastive alignment hurts, but making it too strong also hurts, reinforcing its role as a weak geometric regularizer rather than the main learning objective.

4.5.5 Stability of Results

Exploratory runs that vary optimization details while keeping the architecture and split fixed remain within about ± 0.003 NDCG@20 of the reported result. The qualitative ordering of methods and ablation effects is stable within this envelope.

4.6 Case Study and Computational Cost

We inspect five popular items with at least three KG aspect edges and record softmax slot weights for three users per item. The heatmap is shown in Figure 5; exact weights appear in Appendix Table 13.

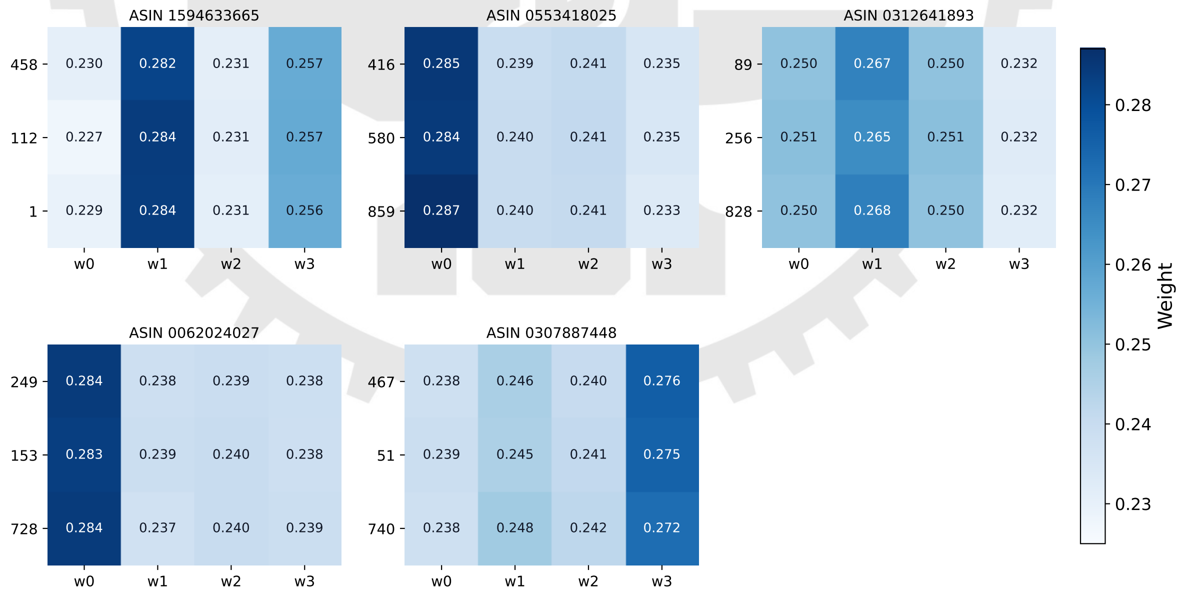

Figure 5: Case-study heatmap of softmax aspect weights for the sampled user–item pairs.

The learned weights are item-conditioned more than user-conditioned. Each sampled item consistently routes mass to a preferred slot, but the three users for the same item produce nearly identical vectors. The dominant weights are only softly peaked, around 0.27–0.29 versus roughly 0.24 on other slots. This matches the side-channel design: because the KG contribution is throttled by the gate, the rationale module learns useful item-level slot routing but not strong per-user slot differentiation.

4.6.1 Computational Cost

RA-GARK trains at about 1.5 seconds per epoch on our hardware, close to LightGCN (1.4 s) and KGRec (1.5 s). The dominant cost is LightGCN propagation; aspect-slot reads, rationale masking, and fusion gates add only a small constant factor at $A = 4$. Full-set inference is also feasible because the item-side rationale computation is vectorized over all $1{,}399$ items.

5 Conclusion and Future Work

5.1 Conclusion

This thesis began from a sparse-KG anomaly: on our review-derived Amazon Books KG, KGAT, KGCL, MCCLK, and KGRec all underperform pure LightGCN. We argued that their shared weakness is mandatory KG fusion. When KG embeddings are forced into the scoring path, sparse or noisy KG signal contaminates the collaborative representation instead of improving it.

RA-GARK addresses this by treating the KG as a gateable side channel. The model keeps a pure LightGCN local view, builds a KG-derived global view from SVD-initialized aspect slots, selects slots with softmax rationale masking, and combines the views through a fusion gate initialized toward the local branch. Three measured components implement the design:

- **KG-SVD initialization**, which preserves aspect-induced semantic geometry and whose removal costs 6.0% NDCG@20.
- **Softmax rationale masking**, whose replacement with sigmoid costs 19.2% NDCG@20.
- **Local-biased fusion**, where $b = +5$ starts the model near pure LightGCN and whose removal costs 3.7% NDCG@20.

With these components, RA-GARK reaches NDCG@20 $= 0.1243$, improving over KGRec by $+13.5\%$ and over LightGCN by $+5.4\%$ at essentially the same per-epoch cost. The main conclusion is that sparse-KG recommendation does not only need better KG aggregation; it needs an architecture that can opt out of KG signal when that signal is unreliable.

5.2 Methodological Insight: Attention Normalization

The largest ablation is the softmax-to-sigmoid replacement in the rationale head. In this architecture, softmax is not mainly useful because it creates one-hot rationale choices; the case study shows only softly peaked weights. Its load-bearing property is scale control: the weights sum to one, so the global item embedding remains bounded before entering a deliberately throttled KG channel. Sigmoid weights, by contrast, allow the KG embedding magnitude to drift with the logits, weakening the calibration provided by the fusion gate.

We state this as a hypothesis, not a universal rule. When a rationale head feeds an auxiliary channel whose contribution is constrained by a fusion gate, sum-to-one normalization may matter more than hard competition. Replication on other datasets and architectures is needed.

5.3 Limitations

Single sparse benchmark All quantitative results come from one Amazon Books subset with 905 users, 1,399 items, and 3,370 KG edges. The motivating anomaly may not hold on denser or cleaner KG benchmarks.

Sparse-KG regime RA-GARK is calibrated for noisy, sparse KG signal. In dense-KG settings, the local-biased "KG off by default" prior may be too conservative, and deep-fusion methods may be preferable.

KG construction source The item–aspect KG is adopted from [19]. This thesis changes the recommender architecture, not the KG extraction pipeline.

5.4 Future Work

Multi-dataset replication The attention-normalization finding should be tested on additional rationale-aware recommenders and item-attribute schemas by changing only the normalization head while holding the rest of the model fixed.

Dense-KG validation RA-GARK should be compared with KG-aware baselines on dense, curated KGs. This would clarify whether gated late fusion remains competitive when KG signal is reliable.

Cross-domain gating The local-biased fusion principle may transfer to other settings with one robust signal and one high-variance auxiliary signal, such as multimodal recommendation or cold-start side information.

Reverse-sparse regime When collaborative interactions are sparse but the KG is dense, the safe default should likely flip: the KG channel should be open at initialization, and the collaborative branch should be treated as the uncertain side. Designing that variant requires changing the gate bias and possibly the role of KG-SVD.

References

[1] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, M. Wang, LightGCN: Simplifying and powering graph convolution network for recommendation, in: Proc. ACM SIGIR, 2020, pp. 639–648.

[2] X. Wang, X. He, M. Wang, F. Feng, T.-S. Chua, Neural graph collaborative filtering, in: Proc. ACM SIGIR, 2019, pp. 165–174.

[3] J. Wu, X. Wang, F. Feng, X. He, L. Chen, J. Lian, X. Xie, Self-supervised graph learning for recommendation, in: Proc. ACM SIGIR, 2021, pp. 726–735.

[4] X. Wang, X. He, Y. Cao, M. Liu, T.-S. Chua, KGAT: Knowledge graph attention network for recommendation, in: Proc. ACM SIGKDD, 2019, pp. 950–958.

[5] Y. Yang, C. Huang, L. Xia, C. Li, Knowledge graph contrastive learning for recommendation, in: Proc. ACM SIGIR, 2022, pp. 1434–1443.

[6] D. Zou, W. Wei, X.-L. Mao, Z. Wang, M. Qiu, F. Zhu, X. Cao, Multi-level cross-view contrastive learning for knowledge-aware recommender system, in: Proc. ACM SIGIR, 2022, pp. 1358–1368.

[7] Y. Yang, C. Huang, L. Xia, C. Huang, Knowledge graph self-supervised rationalization for recommendation, in: Proc. ACM SIGKDD, 2023, pp. 3046–3056.

[8] X. Ren, L. Xia, J. Zhao, D. Yin, C. Huang, Disentangled contrastive collaborative filtering, in: Proc. ACM SIGIR, 2023, pp. 1137–1146.

[9] A. van den Oord, Y. Li, O. Vinyals, Representation learning with contrastive predictive coding, arXiv preprint arXiv:1807.03748 (2018).

[10] R. K. Srivastava, K. Greff, J. Schmidhuber, Highway networks, arXiv preprint arXiv:1505.00387 (2015).

[11] J. Ma, Z. Zhao, X. Yi, J. Chen, L. Hong, E. H. Chi, Modeling task relationships in multi-task learning with multi-gate mixture-of-experts, in: Proc. ACM SIGKDD, 2018, pp. 1930–1939.

[12] H. Tang, J. Liu, M. Zhao, X. Gong, Progressive layered extraction (PLE): A novel multi-task learning (MTL) model for personalized recommendations, in: Proc. ACM RecSys, 2020, pp. 269–278.

[13] Z. Wang, G. Lin, H. Tan, Q. Chen, X. Liu, CKAN: Collaborative knowledge-aware attentive network for recommender systems, in: Proc. ACM SIGIR, 2020, pp. 219–228.

[14] H. Wang, F. Zhang, M. Zhao, W. Li, X. Xie, M. Guo, Multi-task feature learning for knowledge graph enhanced recommendation, in: Proc. The Web Conference (WWW), 2019, pp. 2000–2010.

[15] X. Wang, T. Huang, D. Wang, Y. Yuan, Z. Liu, X. He, T.-S. Chua, Learning intents behind interactions with knowledge graph for recommendation, in: Proc. The Web Conference (WWW), 2021, pp. 878–887.

[16] G. Zhou, X. Zhu, C. Song, Y. Fan, H. Zhu, X. Ma, Y. Yan, J. Jin, H. Li, K. Gai, Deep interest network for click-through rate prediction, in: Proc. ACM SIGKDD, 2018, pp. 1059–1068.

[17] X. He, Z. He, J. Song, Z. Liu, Y.-G. Jiang, T.-S. Chua, NAIS: Neural attentive item similarity model for recommendation, IEEE Transactions on Knowledge and Data Engineering 30 (12) (2018) 2354 –2366.

[18] J. Xiao, H. Ye, X. He, H. Zhang, F. Wu, T.-S. Chua, Attentional factorization machines: Learning the weight of feature interactions via attention networks, in: Proc. International Joint Conference on Artificial Intelligence (IJCAI), 2017, pp. 3119–3125.

[19] I.-N. Ho, Llm-enhanced multimodal product recommendation with review aspect-specific knowledge graphs, Master's thesis, Institute of Information Management, National Yang Ming Chiao Tung University (2024).

Appendix

Table 13 lists the softmax aspect weights used to draw the case-study heatmap in Chapter 4. The main text keeps the visual summary and interpretation, while this appendix preserves the exact numeric values.

Table 13: Softmax aspect weights on sampled user–item pairs.

Item (asin)	Top KG aspects	User	w_0	w_1	w_2	w_3
1594633665	character_complexity, unreliable_narrators, psychological_themes	458	0.230	**0.282**	0.231	0.257
		112	0.227	**0.284**	0.231	0.257
		1	0.229	**0.284**	0.231	0.256
0553418025	science_fiction_genre, survival_stories, space_exploration	416	**0.285**	0.239	0.241	0.235
		580	**0.284**	0.240	0.241	0.235
		859	**0.287**	0.240	0.241	0.233
0312641893	cyberpunk_genre, cyborg_elements, future_setting	89	0.250	**0.267**	0.250	0.232
		256	0.251	**0.265**	0.251	0.232
		828	0.250	**0.268**	0.250	0.232
0062024027	dystopian_worlds, post_apocalyptic_settings, action_and_suspense	249	**0.284**	0.238	0.239	0.238
		153	**0.283**	0.239	0.240	0.238
		728	**0.284**	0.237	0.240	0.239
0307887448	dystopian_sci_fi, video_games, 80s_culture	467	0.238	0.246	0.240	**0.276**
		51	0.239	0.245	0.241	**0.275**
		740	0.238	0.248	0.242	**0.272**